Conditioning Physics-Informed Parametric Surrogates on the Full Problem Specification

Anonymous Author(s)
Affiliation
Address
email

Abstract

Parametric physics-informed surrogates such as P^{2}INN amortize training over PDE coefficients by encoding them into a latent code. The initial condition (IC) and boundary condition (BC) enter only through the loss, so a trained model cannot be queried at a new IC or BC. We turn the coefficient encoder into a *specification encoder* that reads coefficients, ICs and BCs. The decoder, the concatenation interface, the physics loss and the training budget stay fixed. We then ask, in a pre-registered study with fresh confirmatory test sets, which encoding lets one physics-trained model generalize. We use the logistic reaction family, whose solution depends nonlinearly on the IC. Every IC-conditioned encoder trained on 130 ICs is 4–16$\times$ more accurate than any coefficient-only surrogate on unseen IC parameters, unseen IC families and unseen coefficient–IC pairings. The encoders differ in what they do best. Our pre-registered lift–project encoder (a learned pointwise lift followed by a fixed Fourier projection) is the only one that keeps the original single-IC accuracy, and it is the most accurate on unseen pairings. It does not, however, beat raw-sample conditioning on unseen families. A *factorized* encoder, in which a dedicated raw-sample IC branch is added to the coefficient code, does beat it. The win replicates on six never-used IC families, carries over to convection with boundary-condition conditioning and to a joint two-family model, while combining both representations in one branch makes things worse. Oracle codes explain this. The frozen decoder already represents held-out solutions about as accurately as seen ones, so out-of-family error is an encoder *amortization gap* that grows with distance in code space. Refining only the 50-dimensional code against the physics loss at test time closes 73–85% of it. We map two generalization boundaries that persist under a doubled budget. Unseen IC frequencies break every encoder because the shared decoder never learned them; added training coverage moves this boundary. Sharp ICs break the Fourier-feature codes. That is an encoder failure, and contrary to our pre-stated explanation, a wider band makes it worse. We also extend the encoder to periodic and inflow-Dirichlet BCs, using an authored and validated reference solver, and to a joint convection–reaction model. A held-out PDE type does not transfer zero-shot.

1 Introduction

Design optimization, uncertainty quantification and inverse problems query the same PDE many times. Between queries the *problem specification* changes: the initial condition (IC), the boundary condition (BC), and the coefficients. Physics-informed neural networks [Raissi et al., 2019] learn from the PDE alone, but each one solves a single specification. P^{2}INN [Cho et al., 2024] amortizes training over coefficients. It encodes the coefficients $\boldsymbol{\mu}$ into a latent code, concatenates that code with coordinate features, and feeds both to a shared decoder. This also overcomes the convection and

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reaction failure modes [Krishnapriyan et al., 2021]. The IC and BC, however, enter only through the training loss, so the trained model cannot be queried at a new IC or BC.

We ask whether the coefficient encoder can become a *specification encoder* that reads $(\boldsymbol{\mu}, u_0, \mathrm{BC})$, while the decoder, the injection interface, the physics loss and the budget stay fixed. Such a model would answer queries at unseen ICs, including unseen IC *families*, at unseen coefficient–IC pairings, at unseen BCs, and possibly across PDE types. This raises two questions:

- Does IC conditioning generalize, or does it only memorize the training ICs?
- Does *structure* in the encoder matter, compared with appending raw IC samples as DeepONet branch networks do [Lu et al., 2021, Wang et al., 2021]?

We answer both with a pre-registered protocol. The headline study uses the logistic reaction family, whose solution depends nonlinearly on the IC. It has 130 training ICs and held-out sets of unseen IC parameters, unseen IC families and unseen pairings. Confirmatory sets cover six more families that had never been used, and each was evaluated exactly once.

Findings.

1. **Specification conditioning works.** Every IC-conditioned encoder is 4–16$\times$ more accurate than a coefficient-only surrogate on every held-out axis. The IC input is genuinely used. None of the 25 IC-conditioned runs of the main study collapsed, whereas one of five coefficient-only reference seeds did.
2. **Factorization is the structure that generalizes. Our pre-registered spectral proposal is not.** Our proposal, LIFTSPEC, is a lift–project encoder: a learned pointwise lift followed by a fixed Fourier projection. It keeps the original single-IC accuracy and is best on unseen pairings, but on unseen families it does not beat raw samples. The pre-registered *factorized* ablation adds a dedicated raw-sample IC branch to the coefficient code. It wins on unseen parameters and families. The win replicates on six never-used families and holds on every held-out axis of a BC study and of a joint two-family study. Feeding both representations to one branch is worse; we report this as a falsified follow-up.
3. **The error is an amortization gap.** Oracle codes show that the frozen decoder already represents held-out solutions about as well as seen ones, so the out-of-family error comes from where the encoder places the code. Refining only the code against the physics loss closes 73–85% of the gap.
4. **Boundaries and limits.** Two boundaries persist under doubled encoder width, epochs and collocation. Unseen frequencies break every encoder, and not even oracle codes fix this: it is a coverage limit of the shared decoder, which added training ICs move. Sharp features break the Fourier-feature code early. That is an encoder failure, but not the band-limit failure we pre-stated. A held-out PDE type does not transfer zero-shot; oracle codes show that this, too, is an encoding failure.

2 Problem setting and specification encoders

Specifications. We study the one-dimensional convection–diffusion–reaction (CDR) family used to benchmark parametric PINNs [Krishnapriyan et al., 2021, Cho et al., 2024],

$$u_t + \beta u_x - \nu u_{xx} - \rho\, u(1-u) = 0, \qquad x \in [0, 2\pi),\ t \in [0, 1]. \tag{1}$$

A *specification* $s = (\boldsymbol{\mu}, u_0, b)$ has three parts: the coefficients $\boldsymbol{\mu} = (\beta, \nu, \rho)$, which also fix the PDE type (convection is $(\beta, 0, 0)$, reaction is $(0, 0, \rho)$); an IC u_0; and a BC b, which is periodic or inflow-Dirichlet (§7). P^{2}INN [Cho et al., 2024] predicts $\hat{u} = g_{\theta_g}([h_{\mathrm{param}}; h_{\mathrm{coord}}], \boldsymbol{\mu}, x, t)$. Here $h_{\mathrm{param}} = g_{\theta_p}(\boldsymbol{\mu})$ and $h_{\mathrm{coord}} = g_{\theta_c}(x, t)$ are 50-dimensional codes, and g_{θ_g} is a residual ReLU decoder that also receives a linear shift $\texttt{modvec}(\boldsymbol{\mu}, x, t)$. Training uses no solution data. It minimizes

$$\mathcal{L} = \langle r^2 \rangle_\Omega + \langle r^2 \rangle_{t=0} + \langle (\hat{u} - u_0)^2 \rangle_{t=0} + \langle \mathcal{B}[\hat{u}]^2 \rangle_{\partial\Omega}, \tag{2}$$

with unit weights, where r is the residual of (1) and $\mathcal{B}[\hat{u}] = \hat{u}(0, t) - \hat{u}(2\pi, t)$ in the periodic case. The IC and BC appear *only in the loss*, so each trained model is tied to one (u_0, b).

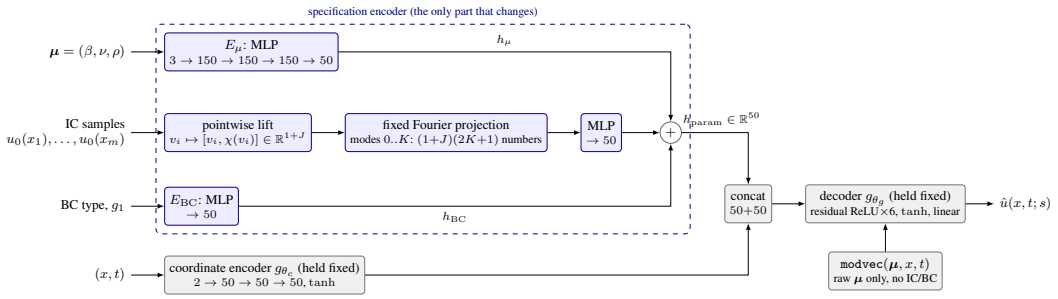


Figure 1: **Specification-conditioned surrogate.** The coordinate encoder, decoder, 50+50 concatenation and `modvec` shift of the coefficient-conditioned P²INN are reused unchanged. Only the parameter-encoder path is extended to read the whole specification $s = (\boldsymbol{\mu}, u_0, \text{BC})$, and it still outputs a 50-dimensional h_{param}. In LIFTSPEC the initial condition is lifted pointwise by a shared scalar map χ and projected onto a fixed Fourier basis (K modes) before a small MLP. Coefficient, IC and BC codes are summed. The raw-sample control instead feeds $[\boldsymbol{\mu}/10, u_0(x_1..x_m), \text{BC}]$ to a single MLP.

Specification-conditioned surrogates. We keep g_{θ_c}, g_{θ_g}, the 50+50 concatenation and `modvec` unchanged; `modvec` still receives only the raw $\boldsymbol{\mu}$. We replace the coefficient encoder with a *specification encoder* $E(s) \in \mathbb{R}^{50}$ (Fig. 1). The loss (2) is unchanged except that each specification uses its own u_0 and $\mathcal{B}$; for inflow, $\mathcal{B}[\hat{u}] = \hat{u}(0, t) - g(t)$. Every encoder sees the IC through the same m=64 equispaced sensor values $v_i = u_0(x_i)$, and every encoder adds the same capacity.

- **Raw-sample conditioning (control).** $E = \text{MLP}([\boldsymbol{\mu}/10,\ v,\ \phi_b])$, where ϕ_b are the BC features and the MLP has the reference encoder's shape ($\cdot \to 150 \to 150 \to 150 \to 50$, tanh). This is the DeepONet branch format [Lu et al., 2021, Wang et al., 2021].
- **LIFTSPEC (pre-registered proposal).**

$$E(s) = E_\mu(\boldsymbol{\mu}/10) + E_{\text{IC}}(v) + E_{\text{BC}}(\phi_b), \qquad E_{\text{IC}}(v) = \text{MLP}\big(\mathcal{F}_K[\,v, \chi(v)\,]\big). \tag{3}$$

  Here E_μ has the reference encoder's architecture, $\chi : \mathbb{R} \to \mathbb{R}^J$ is a scalar lift ($1 \to 16 \to J$) shared across sensors, and $\mathcal{F}_K$ is a *fixed* real Fourier projection of every channel onto modes $0..K$. The final one-hidden-layer MLP is sized so that the encoder has 63,006 parameters, against 63,050 for the control.
- **Ablations** with the same capacity. *Factorized* uses $E_\mu + \text{MLP}(v) + E_{\text{BC}}$, i.e. raw samples in a separate, additively fused branch. *Spectral* uses $E_\mu + \text{MLP}(\mathcal{F}_K v) + E_{\text{BC}}$, i.e. no lift.

Why lift-then-project? CDR solution operators compose pointwise maps, such as the reaction flow $R_t(v) = \sigma(\rho t + \text{logit}\, v)$, with Fourier multipliers, namely convection and diffusion. For reaction, the Fourier coefficients of the exact solution are $\mathcal{F}_k[R_t \circ u_0]$, a pointwise lift followed by a fixed projection. This is exactly the form of (3). We expected this structure to generalize because χ acts on scalars, so an IC from a new family presents it only familiar values, and because no learned weight is tied to a sensor position. The same argument predicted a failure for ICs with structure above mode K. Such ICs do break LIFTSPEC, but §6 shows that the band limit is not the cause.

3 Experimental protocol

Held fixed. We import the reference P²INN code unmodified, and only `param_embed` is produced differently. Our forward pass equals the reference one exactly (maximum difference 0). Residuals come from the reference `PDE_cal`, and points and reference solutions come from the reference generators, matching their outputs to 10^{-15}. Every arm uses the reference budget: Adam with its default learning rate, 20,000 full-batch epochs, 1,000 interior, 256 initial and 100 boundary points per specification, unit loss weights, and seeds 0–4. Errors follow the reference test script. The relative error is $\|\hat{u} - u\|_2 / \|u\|_2$ and the absolute error is the mean of $|\hat{u} - u|$, both on each specification's 1,000 test points, averaged over a set; we report mean±std over seeds. On the authors' released checkpoint, their evaluation script and ours give identical numbers (abs. 0.0042, rel. 0.0090; App. A).

Table 1: **Main results** (relative L_2 error, mean±std over 5 seeds, with the mean absolute error below in gray). "Seen" is the set of training specifications; for the coefficient-only reference it is its own setting. The held-out columns are never trained on. Bold marks the best arm per column. Every IC/BC-conditioned arm has the same budget and the same added encoder capacity (*matched*). The coefficient-only reference is trained in its reference setting (one IC, periodic BC), and on held-out specifications its output is scored as the only answer it can give. In the convection + BC block, "unseen IC family" is under inflow BCs, the pairing column pairs an unseen family with an unseen inflow value, and "unseen BC" is an unseen inflow value g_1. The joint two-family model is reported in Table 9.

Method	Family	Seen spec.	Unseen IC params	Unseen IC family	Unseen μ×IC pairing	Unseen BC	Budget matched?
Coef.-only P^{2}INN (ref. setting)	reaction	0.140±0.272 0.124	0.441±0.198 0.283	0.577±0.169 0.315	0.373±0.306 0.256	–	–
Coef.-only, IC-varied data	reaction	0.289±0.001 0.152	0.260±0.001 0.149	0.378±0.001 0.200	0.161±0.001 0.082	–	n/a
Raw-sample conditioning	reaction	0.028±0.006 0.012	0.052±0.004 0.026	0.106±0.014 0.049	0.021±0.004 0.010	–	yes
factorized, raw	reaction	0.028±0.003 0.012	**0.041±0.003** 0.020	**0.085±0.007** 0.040	0.016±0.001 0.008	–	yes
factorized, spectral	reaction	0.035±0.009 0.017	0.049±0.006 0.025	0.108±0.006 0.051	0.023±0.006 0.012	–	yes
LIFTSPEC (lift–project, pre-registered)	reaction	0.024±0.003 0.010	0.049±0.006 0.025	0.118±0.015 0.055	**0.014±0.002** 0.007	–	yes
Dual pathway (follow-up, Amend. 5)	reaction	**0.024±0.002** 0.011	0.073±0.017 0.038	0.153±0.031 0.072	0.017±0.002 0.009	–	yes
Coef.-only P^{2}INN (ref. setting)	conv. + BC	0.009±0.001 0.005	0.781±0.002 0.376	1.095±0.005 0.380	1.113±0.005 0.387	0.939±0.004 0.362	–
Coef.-only, IC-varied data	conv. + BC	0.556±0.001 0.230	0.557±0.001 0.271	0.665±0.001 0.250	0.689±0.001 0.272	0.535±0.001 0.219	n/a
Raw-sample conditioning	conv. + BC	**0.045±0.002** 0.017	0.231±0.014 0.106	**0.333±0.025** 0.115	**0.332±0.023** 0.114	0.182±0.020 0.067	yes
LIFTSPEC (lift–project, pre-registered)	conv. + BC	0.047±0.003 0.018	**0.209±0.033** 0.093	0.363±0.036 0.127	0.363±0.035 0.127	**0.097±0.020** 0.036	yes

Core study. The headline IC axis must be *nonlinear*. The core study therefore uses reaction, $\boldsymbol{\mu} = (0, 0, \rho)$ with $\rho \in \{1, .., 10\}$, which is P^{2}INN's 1–10 setting. Its exact solution $\sigma(\rho t + \operatorname{logit} u_0)$ is closed-form for *any* IC. The ICs are periodic versions of four zoo families (App. C): **G**aussian bumps (`gauss`, `gauss_pi_2`), **S**inusoids with $k \in \{1, 2\}$ (`sin(x)`, `1+sin(x)`, `sin(2x)`), compact **P**olynomial bumps (`x^2`) and $\tanh$ plaTeaus, **T** (`tanh(x)`). The training set contains `gauss_pi_2` and `gauss` at all ten values of ρ, plus 64 G and 64 S ICs from a scrambled Sobol design [Sobol', 1967]. Block A of these ICs gets three coefficients each, at least one of them above 5; block B gets two coefficients at most 5. This gives 340 specifications. The held-out axes, fixed in a timestamped plan before any comparison and never used for training or selection, are:

(a) *unseen IC parameters*: sixteen G/S ICs, each with one parameter beyond its sampled range (160 specifications);

(b) *unseen IC families*: eight P and eight T ICs (160);

(c) *unseen pairings*: the 64 block-B ICs at $\rho \in \{6..10\}$ (320). Every coefficient and every IC is trained, but never in these pairings.

The *original setting*, `gauss_pi_2`×$\rho \in \{1..10\}$, is used to check non-regression.

Arms, selection and decision rule. We compare three kinds of arm. The first is the coefficient-only P^{2}INN in its reference setting, which on a held-out specification can only return its training-IC solution. The second is the same architecture trained on our IC-varied data, which is *IC-blind*. The third is the matched-capacity encoders of §2 (Table 4). For LIFTSPEC, $K \in \{6, 10\}$ and $J \in \{2, 6\}$ were selected on 16 *in-distribution* validation ICs scored on the reference validation points, giving K=6, J=6 (Table 3). A *clear win* over an arm R on an axis requires three things: a margin of at least $\delta_R = \max(0.01, 0.1\, e_R)$, a margin above the pooled seed std, and a paired Wilcoxon test over specifications with $p<0.01$. Non-regression requires every coefficient of the original setting to stay within $\max(1.1\, e_{\text{ref}}, e_{\text{ref}}+0.005)$. Confirmatory sets were pre-registered as dated amendments before their single evaluation.

Table 2: **Out-of-family generalization replicates on never-used families.** Relative L_2 (mean over 5 seeds) on the round-1 held-out axes and on the two confirmatory sets. Those sets were defined after round 1 and evaluated once, on the frozen round-1 checkpoints. A dagger marks a pre-registered clear win over raw-sample conditioning (margin $\geq \delta$, larger than the seed std, Wilcoxon $p < 0.01$). The bold entry is the lowest error in its column.

	Round 1 (P, T)		Fresh set 1 (L, W, M)		Fresh set 2 (H, K, Z)	
Encoder	params	families	params	families	params	families
Coef.-only, IC-varied	0.260	0.378	0.267	0.317	0.268	0.334
Raw-sample conditioning	0.052	0.106	0.069	0.097	0.064	0.096
Factorized, raw	**0.041**†	**0.085**†	**0.048**†	**0.078**†	**0.045**†	**0.076**†
Factorized, spectral	0.049	0.108	0.057†	0.121	0.055	0.107
LIFTSPEC (pre-registered)	0.049	0.118	0.059†	0.135	0.055	0.110
Dual pathway (Amend. 5)	0.073	0.153	0.086	0.153	0.079	0.151

4 Results: one physics-trained surrogate across coefficients and initial conditions

The reference reproduces and is fragile. Four of the five seeds of the coefficient-only P^{2}INN reach 0.0035 in their own setting. The paper reports 0.0089 ± 0.0019, and we also reproduce its other settings (App. A). Seed 0 stays on the trivial-attractor plateau of the reaction failure mode. After epoch 4,000 its IC loss stays between 0.22 and 0.27, and its error for $\rho \geq 4$ is 0.97. Table 1 includes this seed; below we also quote the healthy seeds alone. None of the 25 IC-conditioned runs in Table 1 collapsed; one enlarged-budget diagnostic run did (App. F).

Conditioning on the IC works. A coefficient-only model can only return its training-IC solution. Even its healthy seeds therefore reach just 0.341, 0.493 and 0.220 on axes (a), (b) and (c). The IC-blind model reaches 0.3782 on unseen families. Every IC-conditioned encoder is 4–16$\times$ more accurate on every axis. The errors are 0.041–0.052 on unseen parameters, 0.085–0.118 on unseen families (polynomial bumps and plateaus, which never appear in training), and 0.014–0.023 on unseen pairings. All of these are clear wins (paired $p < 10^{-27}$, every specification better), and they hold at every ρ (Fig. 2). Replacing the IC at test time sends every model back to the IC-blind level (§5).

Which encoder generalizes? Round 1, as pre-registered. LIFTSPEC does *not* clear the decision rule against raw-sample conditioning on any axis. On (a) its margin is $+0.0033$, against $\delta = 0.0100$. On (b) it is worse, with margin -0.0117. On (c) its margin is $+0.0069$; this gain is consistent ($p = 9.6 \times 10^{-54}$, 98% of specifications better) but below δ. The pre-registered *factorized* ablation does clear the rule, on (a) by 0.0113 ($p = 8.1 \times 10^{-20}$) and on (b) by 0.0212 ($p = 1.8 \times 10^{-20}$). On (c) its margin is 0.0045. The control already reaches 0.0209 there, so the 0.01 floor of δ would require halving its error, and no encoder does.

The factorization result replicates on six never-used families. After round 1 we fixed two confirmatory sets (Table 2). The first contains Lorentzian bumps, tents and bimodal mixtures; the second contains sech^2 bumps, asymmetric Gaussians and sawtooth ramps. Each also includes fresh beyond-range draws, and each was evaluated exactly once, on the frozen round-1 checkpoints. The factorized encoder is the most accurate encoder and a clear win over raw samples in *all six* columns ($p \leq 8 \times 10^{-20}$, 84–92% of specifications better). LIFTSPEC and the spectral ablation are again worse than raw samples on new families. What generalizes is therefore *factorization*, i.e. a dedicated IC branch over full-resolution samples whose code is added to the coefficient code. Spectral lifting does not.

A falsified follow-up. A *dual-pathway* encoder feeds raw samples and lift–project features into one IC branch of matched capacity (Amendment 5). We expected it to combine both strengths; it does not. It has the lowest seen error (0.0235), but it is worse than raw samples on fresh families (0.1511 vs. 0.0957) and on round-1 families (0.1531), and it fails non-regression. A richer input through a narrower bottleneck fits the training ICs better but places codes for unfamiliar ICs worse (§5).

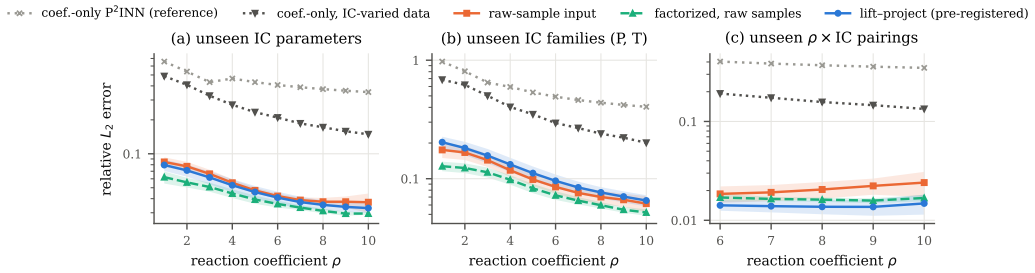


Figure 2: **Per-coefficient error on the held-out axes**: mean over 5 seeds, with ± 1 std bands for the IC-conditioned arms. The coefficient-only reference mean includes one collapsed seed. At every ρ, all IC-conditioned encoders are several times more accurate than the coefficient-only models. The factorized encoder is best on unseen parameters and families, and LIFTSPEC on unseen pairings.

Seen accuracy: only LIFTSPEC keeps the original setting. Adding 128 ICs costs accuracy on the original single-IC problem for every encoder except LIFTSPEC. LIFTSPEC stays within the bound at all ten coefficients (0.0066, against 0.0102 for raw samples and 0.0110 for factorized). It also passes the stricter median-based check (0/10 violations), where the raw, factorized and spectral encoders have 9, 10 and 10 violations.

5 Mechanism: what the encoders represent, and why they generalize or fail

The IC input is used, and the codes are not degenerate. At test time we replace each held-out IC representation in two ways: with the representation of another held-out IC (*shuffled*, a fixed derangement), and with the mean training IC (*constant*). Both destroy every conditioned model. For LIFTSPEC, the error on unseen families rises from 0.118 to 0.449 (shuffled) and 0.379 (constant). The constant result sits at the IC-blind level (0.3782), and all encoders behave alike (Table 5). The codes are also not collapsed. Held-out codes have a participation ratio of 5.5, against 5.2 for training codes, and no two held-out ICs share a code (minimum pairwise distance 0.32 RMS units).

The decoder can represent held-out solutions; the error is an amortization gap. For each held-out specification we fit the best 50-d code for the *frozen* decoder. This *oracle* code is fitted to the reference on the validation points and scored on the test points. On unseen families, oracle codes reach 0.014–0.022 for all encoders, while the encoders' own codes reach 0.095–0.138 (Fig. 3b). A decoder trained only on Gaussian and sinusoid ICs therefore already contains accurate solutions for polynomial bumps and plateaus. The held-out error comes from the encoder *misplacing* the code. It is an amortization gap, not a representational limit. Because the oracle errors are small for every encoder, the encoders differ mainly in where they place codes for unfamiliar ICs. The gap grows with distance from the training codes. Across held-out ICs, the Spearman correlation between error and code distance to the nearest training IC is 0.85 for LIFTSPEC, 0.72 for the factorized encoder and 0.56 for raw samples (Fig. 3a).

Why Fourier-feature codes extrapolate worse. LIFTSPEC sees an IC only through $K{=}6$ lift–project features. In distribution this inductive bias pays off: LIFTSPEC has the lowest seen and pairing errors and is the only encoder that does not regress. Out of family its codes are placed worse, and three diagnostics show this. First, a linear probe from held-out spectral codes back to the IC's Fourier magnitudes fails ($R^2 = -0.36$ for LIFTSPEC and $R^2 = -0.26$ for the spectral ablation), whereas factorized codes remain decodable ($R^2{=}0.24$). Second, the spectral codes are the most input-sensitive (Jacobian norms 15.3 and 9.8, against 3.9), so unfamiliar ICs move them far from the training codes. Third, LIFTSPEC degrades fastest as ICs sharpen, and a wider band makes this worse rather than better (B2 in §6). The cause is therefore not the information discarded above mode K. The learned lift itself is strongly nonlinear: an affine fit explains only $R^2{=}0.880$ of its channels, a logit-type basis $\{1, v, \log v, \log(1{-}v)\}$ explains 0.990, and a cubic explains 0.997 (Fig. 3c). It does build nonlinear value features, but the data do not single out the logistic coordinate.

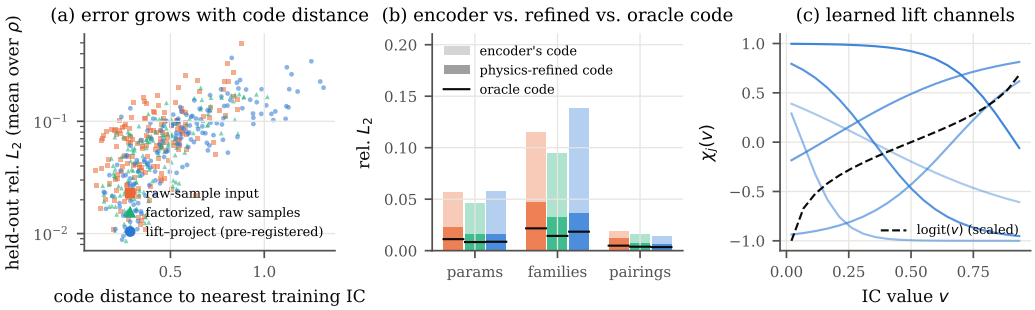


Figure 3: **Mechanism. (a)** Held-out error, averaged over ρ, against the code distance of each held-out IC to its nearest training IC; one point per IC and seed. **(b)** Error with the encoder's code (light), after physics-only refinement of the code (dark), and with the oracle code (black tick), on seeds 0–2 and $\rho \in \{1, 3, 5, 8\}$. **(c)** The learned lift channels χ_j of LIFTSPEC (blue) against a scaled $\mathrm{logit}(v)$.

Physics-only code refinement closes most of the gap. The PDE, the IC and the BC are all known at query time, so the code can be refined without any solution data. We minimize the specification's own loss (2) over the 50-d code alone, for 500 Adam steps from the encoder's output, with all weights frozen. On unseen families this closes 73–85% of the encoder–oracle gap. For example, LIFTSPEC goes from 0.138 to 0.037, and raw samples go from 0.115 to 0.047. On unseen parameters the step closes 75–84%. Refinement is not zero-shot, but it is a cheap mitigation that starts a physics-consistent solve from the amortized code.

6 Out-of-distribution behaviour and generalization boundaries

We move one IC parameter at a time (Gaussian width, sinusoid frequency, amplitude, floor, plateau edge) from inside the trained range to far outside it, and evaluate every arm at $\rho \in \{1..10\}$ against closed-form references (Fig. 4). Amplitudes and floors degrade every encoder gradually (App. F). The other sweeps reveal two boundaries, which we test with pre-stated diagnostics.

B1 – frequency coverage (all encoders). Training contains sinusoids with $k \in \{1, 2\}$. Every IC-conditioned encoder is accurate at k=2 (0.035 for raw samples, 0.037 for LIFTSPEC) but breaks at k=3 (0.235, 0.228 and 0.326 for raw, factorized and LIFTSPEC), no better than the IC-blind model (0.226). Since k=3 lies well inside the K=6 band of LIFTSPEC, this is not truncation. Three pre-stated tests (Table 7, App. F) place the failure in the *coverage* of the shared decoder. (i) It is not under-training. With 2× encoder width, epochs and collocation, the error at $k \geq 3$ stays at 0.377 (LIFTSPEC) and 0.233 (raw samples). (ii) No code fixes it. Even the best code for the frozen decoder (the oracle) errs by 0.183 and 0.170 at $k \in \{3, 4\}$, against 0.007 and 0.012 in range. (iii) Coverage moves it. Adding 32 training sinusoids with $k \in \{3, 4\}$ cuts the error there by 2.6× (LIFTSPEC) and 2.1× (raw samples), with every seed separated. The points at $k \in \{5, 6\}$ keep failing, and the core held-out errors change by at most 0.004.

B2 – sharp ICs (specific to Fourier-feature codes). Sharpening the plateau edge pushes IC energy above mode 6, from 0.0% at e=0.5 to 4.5% at e=0.05. Over that range LIFTSPEC degrades twice as fast as the encoders that see full-resolution samples (0.357 against 0.181 and 0.178). The spectral ablation without a lift shows a smaller excess (0.271). Narrowing the Gaussian behaves the same way (Fig. 4). The excess survives the doubled budget: at both budgets every LIFTSPEC seed is worse than every raw-sample seed. It is an encoder failure. Oracle codes reach 0.131 and 0.090, no worse than for the decoder trained with raw samples (0.141, 0.103), against 0.549 and 0.583 for the codes of LIFTSPEC. It is *not* the band-limit failure we pre-stated. At K=24 these ICs keep at most 0.1% of their energy outside the band, yet the excess over raw samples does not shrink, whether capacity is matched or the hidden width of K=6 is kept (Table 7). A post-hoc check points to code placement instead. Sharp ICs put Fourier-feature codes 1.5–2.6× as far from the nearest training code as raw-sample codes (1.37 against 0.53 training spreads for plateaus). This fits their larger input Jacobians (§5), and the error grows with this distance (Spearman 0.89 over all sweep points).

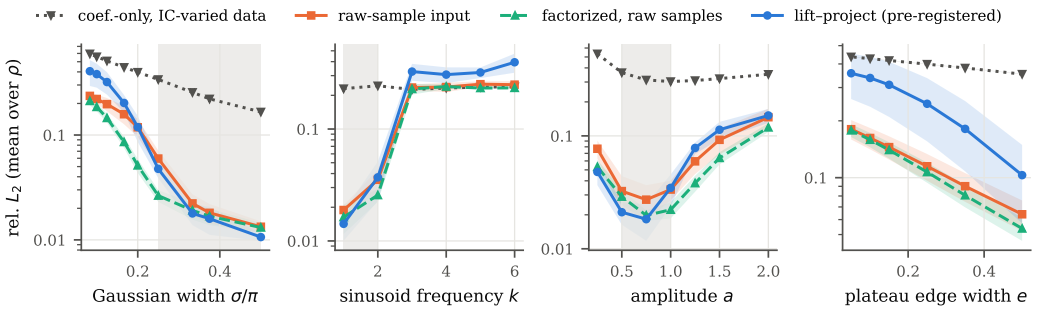


Figure 4: **Out-of-distribution sweeps** on the reaction family (mean over ρ and 5 seeds, ± 1 std; gray: trained range). The frequency sweep shows a cliff shared by every encoder (B1); the width and edge sweeps show the extra failure of LIFTSPEC on sharp ICs (B2).

7 Boundary conditions, a second PDE family, and a held-out PDE type

Conditioning on the boundary condition (convection). For convection, $u_t + \beta u_x = 0$ with $\beta \in \{1..5\}$, one model is conditioned on two BC types. The first is *periodic*, with the reference boundary term. The second is *inflow-Dirichlet*, $u(0,t) = g(t)$, where g relaxes from $u_0(0)$ to a value g_1; its loss term is $\hat{u}(0,t) - g(t)$. The BC enters as a one-hot type together with g_1. In the structured encoders it goes through E_{BC}, and the raw-sample control appends it to its input. Inflow references come from our method-of-lines solver (App. B). It agrees with the spectral reference on periodic problems to 9.2×10^{-5} and with the exact characteristic solution to 1.7×10^{-5}. Training uses $g_1 \in \{0.1, 0.5, 0.9\}$. The values $g_1 \in \{0.3, 0.7\}$ are held out (Table 8). The BC is genuinely used. Giving inflow problems periodic BC features at test time raises the error of LIFTSPEC on held-out BC configurations from 0.097 to 0.247, and that of raw-sample conditioning from 0.182 to 0.275. On held-out BC configurations, both structured encoders clearly beat raw-sample conditioning (0.1818): the factorized encoder reaches 0.0822 (p=2.8×10^{-28}) and LIFTSPEC reaches 0.0969 (p=2.6×10^{-28}). The factorization result carries over to convection. The factorized encoder clearly beats raw-sample conditioning on 5 of the 5 axes of this study ($p \leq 5 \times 10^{-15}$). Examples are unseen families under inflow BCs (0.2092 vs. 0.3328) and an unseen family with an unseen BC value (0.2118 vs. 0.3319). LIFTSPEC is again worse than raw samples on unseen families (0.3630). Errors are higher than for reaction, since each IC appears in only two training specifications. Spanning both BC types also costs accuracy on the original periodic problem for every conditioned encoder (0.0232 vs. 0.0088 for the reference), so non-regression fails in this study.

Two PDE families in one model. One model is trained on convection ($\beta \in \{1..5\}$, both BC types) and reaction ($\rho \in \{1..5\}$, periodic). The PDE type is carried by $\boldsymbol{\mu}$, and each IC appears once per family (Table 9). The model solves both families on seen problems: 0.0352 and 0.0167 on the original convection and reaction problems for LIFTSPEC. The factorized encoder again clearly beats raw-sample conditioning on every held-out axis within both families, and on the held-out composition of convection, an unseen family and an unseen inflow value. Unseen convection families reach 0.2878 against 0.4139; unseen reaction families 0.1812 against 0.2512; the composition 0.2611 against 0.3980. All have $p \leq 1 \times 10^{-13}$. *Held-out PDE type.* Convection–reaction ($\beta, \rho > 0$ together) appears in no form during training. Zero-shot, every encoder lands at 0.321–0.346. That is the level of the *exact* pure-convection or pure-reaction solution (0.317 and 0.377), so the models do not compose the two mechanisms. Two further pre-registered attempts locate the failure (App. G). A term-wise coefficient branch $e_\beta + e_\nu + e_\rho$ does not help (0.343). Oracle codes for the frozen joint decoder reach 0.070, against 0.310 for the encoder's code, so what fails is the *encoding* of a never-seen combination of terms. Physics-only refinement recovers less of this gap than for ICs (0.233). We report the held-out type as a negative result. Table 10 summarizes what one model covers.

8 Related work

Parametric physics-informed surrogates. PINN training [Raissi et al., 2019] has been amortized over problem families in several ways: parameters as extra inputs, hypernetworks [de Avila Belbute-Peres et al., 2021, Cho et al., 2023], meta-learned initializations [Penwarden et al., 2023], reduced bases [Chen and Koohy, 2024], and latent codes that are auto-decoded and adapted at test time [Huang et al., 2022] or evolved by latent dynamics [Wang et al., 2024, Liang et al., 2026]. P^2INN [Cho et al., 2024] showed that a dedicated coefficient encoder overcomes the convection and reaction failure modes [Krishnapriyan et al., 2021]. These works condition on coefficients. When ICs vary, they either infer a latent by test-time optimization [Huang et al., 2022] or keep the IC fixed. We keep the P^2INN decoder, interface, loss and budget, and ask which *encoder* lets one physics-trained model be queried at new ICs and BCs without optimization. Our physics-only code refinement links this amortized encoder back to MAD-style adaptation.

Operator learning and function encoders. DeepONet and physics-informed DeepONet [Lu et al., 2021, Wang et al., 2021] encode input functions by raw sensor values, as our control does. FNO and neural operators [Li et al., 2021, Kovachki et al., 2023, Li et al., 2024] lift pointwise and then act spectrally. They are the closest relatives of our lift–project encoder, which we use only to produce a global code. PCA-Net [Bhattacharya et al., 2021] projects inputs onto a fixed basis. PINTO [Boya and Subramani, 2024] conditions on ICs and BCs through coordinate-dependent cross-attention. Neural fields obtain global codes by auto-decoding [Park et al., 2019, Dupont et al., 2022, Serrano et al., 2023, Yin et al., 2023], from set encoders [Garnelo et al., 2018, Zaheer et al., 2017], or by projection onto learned bases [Ingebrand et al., 2024]. Multiplicative FiLM conditioning [Perez et al., 2018] is excluded here by the fixed interface. Our oracle-code analysis separates decoder representability from the encoder's amortization gap.

Multi-physics conditioning. PDEformer [Ye et al., 2024], Unisolver [Zhou et al., 2025] and CAPE [Takamoto et al., 2023] condition data-driven models on descriptions of the equation, IC and BC, or on coefficients. In-context and foundation models [Yang et al., 2023, Herde et al., 2024, McCabe et al., 2024, Serrano et al., 2024] condition on trajectories, and BOON [Saad et al., 2023] enforces BCs in operator outputs. All of these train on simulated solutions. Our models use only the residual, IC and BC terms.

9 Limitations and conclusion

Limitations.

- Our pre-registered proposal did not meet our own decision rule against raw-sample conditioning. The factorized encoder that did entered as an ablation and was confirmed on sets defined after round 1. Every round is reported.
- No encoder reached the 0.01 floor on unseen pairings, where the control's error is already 0.0209.
- The study is one-dimensional, each IC family has three to five parameters, and the convection and BC studies have sparser IC coverage. Every conditioned encoder regresses on the original periodic convection problem.
- Decoder, interface and budget are fixed by design, so our conclusions concern the encoder. The frequency boundary lies in the shared decoder, and only added training coverage moved it.

Conclusion. Changing only the parameter encoder turns a coefficient-conditioned physics-informed surrogate into a specification-conditioned one. It answers queries at unseen ICs, IC families, pairings and BC values several times more accurately than coefficient-only models. *How* the specification is encoded matters, but not as we pre-registered. Band-limited lift–project codes are best in distribution, while factorized full-resolution codes generalize out of family: across six never-used families, under BC conditioning, and across two PDE families. Oracle codes explain the pattern. The shared decoder already represents held-out solutions, and the remaining amortization gap of the encoder grows with code distance; physics-only refinement of the code closes most of it. Three questions remain open: the decoder's frequency coverage, the extrapolation of Fourier-feature codes, and transfer to an unseen PDE type.

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A Fidelity to the reference implementation

- **Architecture.** We instantiate the reference `P2INN_phase1` module itself. Our decoding path copies its forward pass after `param_embed`. With the reference coefficient encoder plugged in, outputs agree with `P2INN_phase1.forward` to a maximum absolute difference of exactly 0 for $\rho \in \{1, 5, 10\}$.
- **Parameter counts.** The reference has 76,851 parameters: 53,450 in the coefficient encoder and 23,401 in the coordinate encoder, decoder and `modvec`. The last 23,401 are identical in every arm.
- **Data.** Collocation, initial, boundary and test points and their reference values match the CSV files written by the unmodified `gen_reac.py` and `gen_conv.py` to 8.9×10^{-16}, for coefficients 1–10.
- **Training.** From the same seed, the authors' unmodified `train.py` and our harness give identical first-epoch losses (5.99640 and 5.99640). Over the first 50 epochs they agree within 0.07%. After that they drift apart chaotically, because the reference re-shuffles its full batch every epoch and so changes the floating-point summation order.
- **Evaluation.** On the authors' released checkpoint (reaction, coefficients 1–20, epoch 10,000), the authors' `test_single.py` reports abs. 0.0042 and rel. 0.0090. Our evaluator reports 0.0042 and 0.0090. The paper's Table 2 reports 0.0042 and 0.0092 for this setting.
- **Reproduced reference settings** (coefficient-only P^{2}INN, `gauss_pi_2`, 5 seeds, our harness, relative error):
  - reaction 1–5: 0.0032 ± 0.0012 (paper 0.0027 ± 0.0030);
  - reaction 1–10: 0.0035 over the four non-collapsed seeds (paper 0.0089 ± 0.0019);
  - convection 1–5: 0.0088 ± 0.0013 (paper 0.0079 ± 0.0008);
  - convection 1–10: 0.0140 ± 0.0038 (paper 0.0179 ± 0.0072).
- **Verbatim long runs.** The reference data loader costs about 0.9 s per epoch, so a 20,000-epoch run takes about 5 h. That exceeds our job time limit, so full-length runs of the verbatim scripts were not possible (`attempts_log.md`, A1). All baseline numbers in this paper come from 20,000-epoch runs of the same model, loss and optimizer in our harness.
- **Implementation details that do not change the computation.** A CUDA graph captures the training step. Adam is used in its `capturable` form, which is the same algorithm with the default learning rate. Specification codes are repeated per point by `expand`, which gives the same values as indexing. On GPU, repeating a run with the same seed changes results by about 10^{-3}, because cuBLAS split-K reductions are not bitwise deterministic.

B Reference solutions and their validation

- **Reaction.** The reference `reaction_solution` is the closed-form logistic solution. It is exact for any initial condition, including every held-out IC and every out-of-distribution sweep.
- **Periodic convection.** The reference `convection_diffusion_discrete_solution` is the exact spectral solution.
- **Inflow-Dirichlet convection (authored).** We wrote a method-of-lines solver: third-order upwind-biased differences, classical RK4 at CFL 0.4, ghost values from the boundary characteristic, and an outflow closure. It is run at $N{=}4096$ and sampled on the $256{\times}100$ reference grid. We validated it in three ways:
  - On periodic specifications, using the same scheme with periodic wrap, it agrees with the spectral reference to a relative L_2 difference of at most 9.2×10^{-5}.
  - Under inflow, it agrees with the exact characteristic solution to at most 1.7×10^{-5}.
  - Refining from $N{=}4096$ to 8192 changes the solution by at most 1.4×10^{-5}, and the median observed order is 1.9.

These figures cover seven ICs from all four families, $\beta \in \{1, 3, 5\}$ and five inflow values. The inflow data $g(t) = g_1 + (A + Bt)e^{-t/\tau}$, with $\tau = 0.1$, are chosen so that $g(0) = u_0(0)$ and $g'(0) = -\beta u_0'(0)$. This makes the corner compatible to first order.

- **Convection–reaction (held-out PDE type).** The reference CDR operator-split solver at ν=0 is used unchanged. Convection and pointwise reaction commute, so the splitting is exact up to discretization:
  - at its own time levels it matches the exact characteristic–logistic solution to 1.4×10^{-4};
  - halving Δt changes it by 1.7×10^{-6};
  - doubling n_x changes it by 1.3×10^{-4}.

  The reference splitters advance in steps of $1/n_t$ but label the outputs on `linspace(0,1,`n_t`)`. Compared at the labelled times, this convention differs from the exact solution by up to 1.8×10^{-2}. That error floor is far below the errors at stake in §7.

C Specification space and splits

All ICs are 2π-periodic. $y = (x - s) \bmod 2\pi$ is a periodic shift of the corresponding construction in the reference IC zoo. Trained families:

- G, Gaussian bumps $c + a\, e^{-(y-\pi)^2/2\sigma^2}$, built like the zoo's `gauss` and `gauss_pi_2`. Amplitude $a \in [0.5, 1]$, offset $c \in [0.02, 0.15]$, width $\sigma \in [\pi/4, \pi/2]$, shift $s \in [0, 2\pi)$.
- S, sinusoids $c + A \sin k(x - s)$. Amplitude $A \in [0.15, 0.45]$, floor $f = c - A \in [0.02, 0.15]$, frequency $k \in \{1, 2\}$, shift s.

Held-out families:

- P, compact polynomial bumps $c + a(1 - ((y - \pi)/w)^2)^2$ on $|y - \pi| < w$. $a \in [0.5, 1]$, $c \in [0.02, 0.15]$, half-width $w \in [\pi/3, 3\pi/4]$.
- T, plateaus $c + \frac{a}{2}[\tanh \frac{y-\pi+w}{e} - \tanh \frac{y-\pi-w}{e}]$. $a \in [0.5, 1]$, $c \in [0.02, 0.15]$, half-width $w \in [0.5, 1.5]$, edge $e \in [0.25, 0.5]$.

Held-out axis (a) moves one parameter beyond its range: G with σ=$\pi/5$ ($\times 4$), σ=0.6π ($\times 2$) or a=1.15 ($\times 2$); S with A=0.5 at floor 0.05, or floor 0.2 (four each, $k \in \{1, 2\}$). A supplementary axis (d), unseen coefficients, crosses $\rho \in \{1.5, \ldots, 9.5\} \cup \{11, 12\}$ with 18 trained ICs. The results are 0.0409 (raw samples), 0.0449 (factorized) and 0.0411 (LIFTSPEC), against 0.4226 for the coefficient-only reference. All ICs stay in $(0, 1.3]$, so the logistic solution is regular. Confirmatory families (Amendment 4):

- L, Lorentzian $c + a/(1 + ((y - \pi)/w)^2)$, $w \in [0.4, 1]$;
- W, tents $c + a \max(0, 1 - |y - \pi|/w)$, $w \in [1, 2.5]$;
- M, bimodal mixtures of two periodic Gaussians with unequal heights.

Second confirmatory set (Amendment 5):

- H, sech^2 bumps;
- K, asymmetric two-sided Gaussians;
- Z, smoothed sawtooth ramps.

The training ICs are 64 G and 64 S draws from scrambled Sobol sequences (seeds 101, 102), plus the zoo's `gauss_pi_2` and `gauss`. Block A, the even-indexed draws, is paired with three coefficients: one from $\{1..5\}$, one from $\{6..10\}$ and one more. Block B, the odd-indexed draws, is paired with two coefficients from $\{1..5\}$. Pairings use RNG seed 7. The exact lists of every set are in `code/specification_encoder_plan*.json`. The plan was frozen at 2026-10-03 17:49 UTC, before any comparison run. Every later change is a dated amendment in `specification_encoder_plan.md`.

D Hyper-parameter selection and budgets

Table 3: **Pre-registered hyper-parameter selection** for LIFTSPEC (seeds 0,1). The criterion is the mean relative error on 16 in-distribution validation ICs. All columns are scored on the reference validation points. No held-out set was evaluated before the selection was frozen.

K	J	Encoder params	Validation ICs (selection)	Original setting	Seen specifications
6	2	63,016	0.0290	0.0063	0.0243
6	6	63,006	0.0262	0.0064	0.0228 ← selected
10	2	63,028	0.0281	0.0085	0.0260
10	6	63,138	0.0378	0.0058	0.0236

Table 4: **Budgets** (from the run logs). Encoder = parameter/specification encoder. Held fixed = coordinate encoder + decoder + `modvec`, identical in every arm. Collocation counts are 1,000 interior, 256 initial and 100 boundary points per specification. Wall-clock is per run, on a shared H200.

Study	Arm	Encoder params	Held-fixed params	Train specs	Interior points	Epochs	Seeds	Min/run
reaction (core)	Coef.-only P^{2}INN (ref. setting)	53,450	23,401	10	10,000	20,000	5	1.7
reaction (core)	Coef.-only, IC-varied data	53,450	23,401	340	340,000	20,000	5	13.2
reaction (core)	Raw-sample conditioning	63,050	23,401	340	340,000	20,000	5	13.3
reaction (core)	factorized, raw	63,045	23,401	340	340,000	20,000	5	13.3
reaction (core)	factorized, spectral	63,036	23,401	340	340,000	20,000	5	13.2
reaction (core)	LIFTSPEC (lift–project, pre-registered)	63,006	23,401	340	340,000	20,000	5	13.3
reaction (core)	Dual pathway (follow-up, Amend. 5)	63,110	23,401	340	340,000	20,000	5	37.2
convection + BC	Coef.-only P^{2}INN (ref. setting)	53,450	23,401	5	5,000	20,000	5	2.9
convection + BC	Coef.-only, IC-varied data	53,450	23,401	296	296,000	20,000	5	24.0
convection + BC	Raw-sample conditioning	63,500	23,401	296	296,000	20,000	5	21.9
convection + BC	factorized, raw	63,499	23,401	296	296,000	20,000	5	26.4
convection + BC	LIFTSPEC (lift–project, pre-registered)	63,494	23,401	296	296,000	20,000	5	23.0
conv. + reac. (joint)	Coef.-only, IC-varied data	53,450	23,401	286	286,000	20,000	3	20.9
conv. + reac. (joint)	Raw-sample conditioning	63,500	23,401	286	286,000	20,000	5	21.5
conv. + reac. (joint)	factorized, raw	63,499	23,401	286	286,000	20,000	5	23.8
conv. + reac. (joint)	LIFTSPEC (lift–project, pre-registered)	63,494	23,401	286	286,000	20,000	5	20.3

E Additional attribution and mechanism results

Table 6 breaks the held-out errors down by family.

- The factorized encoder is the most accurate IC-conditioned encoder on every held-out family.
- The hardest families for every encoder are the bimodal mixtures (M) and the compact polynomial bumps (P). Both put structure where the trained single-bump and sinusoid families have none: a second, unequal bump, and an exactly flat tail with a C^1 edge.
- Fig. 5 shows typical error maps. The error of the encoders' own codes is spread over the whole solution, as a misplaced code would produce. After physics-only refinement of the code, the residual error is small and localized.

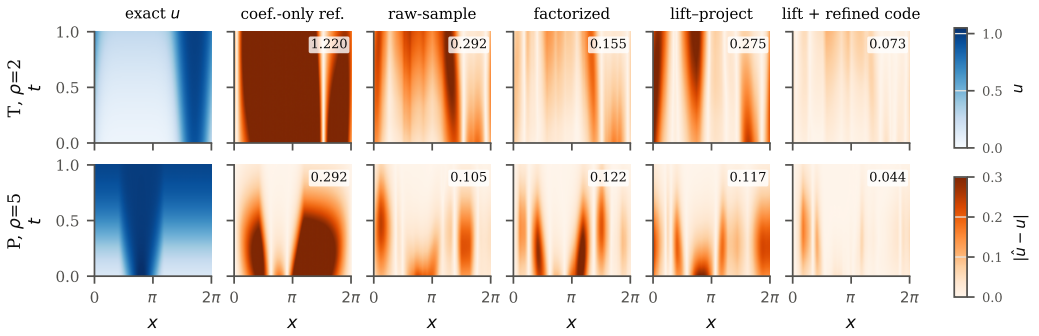


Figure 5: **Held-out-family examples** (seed 1). Rows: a T plateau at ρ=2 and a P bump at ρ=5. Columns: the exact solution, then absolute-error maps of each arm, and of LIFTSPEC after physics-only refinement of its code (500 steps, all weights frozen). Insets give the relative L_2 error on the full 256×100 grid (`logs/reaction_core/solutions_figure.json`).

Table 5: **Attribution and code statistics** (reaction core; mean over 5 seeds). For each held-out axis we report the relative L_2 with the true IC (intact), with the representation of another held-out IC (shuffled), and with the mean training IC (constant). We also give the Spearman correlation between per-IC held-out error and the code distance of the IC to the nearest training IC, the participation ratio (PR) of held-out IC codes, and the R^2 of a ridge probe from codes to the Fourier magnitudes (modes 0–4) of the IC, fitted on training ICs and scored on held-out ICs.

Encoder	(a) intact/shuf./const.	(b) intact/shuf./const.	(c) intact/shuf./const.	Spearman(err, d_{code})	PR	probe R^2
Raw-sample	0.052 / 0.339 / 0.269	0.106 / 0.451 / 0.391	0.021 / 0.219 / 0.168	0.56	4.6	0.28
Factorized, raw	0.041 / 0.347 / 0.259	0.085 / 0.458 / 0.377	0.016 / 0.218 / 0.159	0.72	4.3	0.24
Factorized, spectral	0.049 / 0.347 / 0.260	0.108 / 0.454 / 0.372	0.023 / 0.218 / 0.158	0.76	4.7	-0.26
LIFTSPEC	0.049 / 0.342 / 0.258	0.118 / 0.449 / 0.379	0.014 / 0.219 / 0.161	0.85	5.5	-0.36
Dual pathway	0.073 / 0.344 / 0.260	0.153 / 0.456 / 0.385	0.017 / 0.220 / 0.164	0.78	5.4	-0.06

Table 6: **Per-family relative** L_2 (reaction; mean over 5 seeds, $\rho \in \{1..10\}$ and the ICs of each family). G and S are in-distribution validation ICs of the trained families. All other families are held out: P, T in round 1; L, W, M in the first fresh set; H, K, Z in the second. Bold marks the best IC-conditioned arm per family.

Arm	G	S	P	T	L	W	M	H	K	Z
Coef.-only ref.	0.469	0.500	0.617	0.538	0.490	0.612	0.520	0.625	0.689	0.324
IC-blind	0.263	0.266	0.415	0.342	0.276	0.372	0.304	0.390	0.443	0.168
Raw	0.027	0.040	0.121	0.091	0.073	0.083	0.135	0.094	0.106	0.087
Factorized	0.020	**0.030**	**0.097**	**0.073**	**0.065**	**0.060**	**0.110**	**0.077**	**0.082**	**0.068**
Spectral	0.030	0.035	0.123	0.093	0.109	0.096	0.157	0.117	0.120	0.084
LIFTSPEC	**0.020**	0.034	0.126	0.110	0.134	0.093	0.178	0.116	0.117	0.098
Dual	0.032	0.052	0.174	0.132	0.130	0.130	0.197	0.164	0.168	0.121

F Generalization-boundary diagnostics

Protocol. Each test below was stated in the plan before it was run (Amendments 6, 8 and 9). All use the closed-form logistic references, so every error is the surrogate's.

- **Under-training.** LIFTSPEC and raw-sample conditioning are retrained (seeds 0–2) with every encoder-path width doubled, 40,000 epochs, and 2,000 interior collocation points per specification. The encoders grow from 63,006 to 215,622 parameters (LIFTSPEC) and from 63,050 to 216,050 (raw samples). The collocation points are the reference 1,000 plus as many new grid points, none of which is a test or validation point. The decoder and coordinate encoder are unchanged. A failure *persists* if its error at the failing points stays at least $2/3$ of its standard-budget value.
- **Representability.** For seeds 0 and 1 at $\rho \in \{1, 5\}$, we fit oracle codes to the reference on the validation points and score them on the test points. Physics-refined codes take 500 Adam steps on the specification's own loss, starting from the encoder's code (§5).
- **Coverage.** We add 32 sinusoids with $k \in \{3, 4\}$, each paired with two coefficients, giving 404 training specifications, and retrain both encoders at the standard budget. The core held-out sets still contain none of the added ICs.
- **Band width** (diagnostic only). We retrain LIFTSPEC with $K \in \{12, 24\}$ at matched capacity, and once more at $K{=}24$ with the hidden width of $K{=}6$. $K{=}6$ remains the method, selected on seen-specification validation (Table 3).

Outcomes.

- **Persistence.** The criterion holds at every failing point group for both encoders. The enlarged-to-standard error ratios (LIFTSPEC, raw samples) are 1.21 and 1.00 at $k{\geq}3$, 0.97 and 1.24 at plateau edges $e{\leq}0.1$, and 1.08 and 1.40 at Gaussian widths $\sigma{\leq}\pi/8$. At the enlarged budget the smallest error at any failing point group over all seeds is 0.197. The largest in-range error is 0.049.
- **More budget fits better but generalizes worse.** The seen-specification error falls from 0.022 to 0.017 (LIFTSPEC) and from 0.029 to 0.018 (raw samples). Over the same seeds the unseen-family error rises, from 0.118 to 0.138 and from 0.100 to 0.127. So the held-out errors of the main study are not an under-training artifact either.

Table 7: **Generalization-boundary diagnostics** on the reaction sweeps of Fig. 4 (relative L_2, mean over $\rho \in \{1..10\}$ and seeds 0–2, $\pm$ std over seeds). Columns group the sweep points. The references are closed-form. †One of three LIFTSPEC runs at the enlarged budget did not fit its own training set and is excluded (App. F). ‡Band-width diagnostic: the standard LIFTSPEC has K=6 and hidden width 66 after the projection. At matched capacity, widening the band narrows that layer; the last LIFTSPEC row keeps it at 66, with more parameters.

Arm	Condition	$k \leq 2$ (trained)	$k \in \{3,4\}$	$k \in \{5,6\}$	plateau $e \leq 0.1$	Gaussian $\sigma \leq \pi/8$
LIFTSPEC	standard budget	0.021±0.004	0.295±0.024	0.328±0.030	0.372±0.071	0.390±0.074
	2× width, epochs, collocation†	0.026±0.002	0.336±0.049	0.417±0.142	0.362±0.091	0.421±0.014
	+32 training sinusoids with $k \in \{3,4\}$	0.025±0.003	0.113±0.028	0.436±0.031	0.416±0.191	0.434±0.195
	K=12, hidden width 42‡	0.029±0.002	0.380±0.070	0.453±0.068	0.513±0.078	0.413±0.055
	K=24, hidden width 24‡	0.028±0.004	0.367±0.040	0.437±0.038	0.385±0.115	0.605±0.061
	K=24, hidden width 66‡	0.030±0.018	0.317±0.021	0.413±0.099	0.444±0.209	0.704±0.079
Raw samples	standard budget	0.030±0.004	0.224±0.004	0.243±0.007	0.165±0.025	0.202±0.012
	2× width, epochs, collocation†	0.045±0.006	0.232±0.006	0.234±0.007	0.205±0.010	0.284±0.015
	+32 training sinusoids with $k \in \{3,4\}$	0.039±0.018	0.109±0.007	0.241±0.010	0.179±0.001	0.250±0.057
Codes for the standard-budget decoder (seeds 0,1; $\rho \in \{1,5\}$)						
LIFTSPEC	encoder's code	0.032	0.386	0.478	0.549	0.583
	physics-refined code	0.012	0.201	0.294	0.165	0.128
	oracle code	0.007	0.183	0.275	0.131	0.090
Raw samples	encoder's code	0.040	0.313	0.331	0.196	0.281
	physics-refined code	0.024	0.206	0.280	0.177	0.179
	oracle code	0.012	0.170	0.261	0.141	0.103
Post hoc: distance from the IC code to the nearest training-IC code (units of the training-code spread, §5; standard budget, 5 seeds)						
Code distance	raw samples	0.19	0.56	0.59	0.53	0.75
	factorized, raw	0.19	0.59	0.52	0.65	0.74
	factorized, spectral	0.20	1.96	1.85	1.11	1.14
	LIFTSPEC	0.23	1.46	1.90	1.37	1.26
IC energy	above mode 6 (%)	0.0	0.0	0.0	3.5	0.8
	above mode 12 (%)	0.0	0.0	0.0	0.8	0.0
	above mode 24 (%)	0.0	0.0	0.0	0.1	0.0

- **One collapsed run.** One of the three enlarged-budget LIFTSPEC runs (seed 2) did not fit its own training set: its mean seen error is 0.32, while no other run in these diagnostics exceeds 0.040. We exclude runs whose mean seen error exceeds 0.1, a rule written after this collapse was observed. Including the run raises the enlarged-budget errors of LIFTSPEC at the failing points, to 0.475 ($k \geq 3$), 0.478 (plateaus) and 0.564 (Gaussians), so no conclusion depends on the exclusion.
- **Coverage.** Besides moving B1, the added sinusoids change the core held-out errors (unseen parameters, families and pairings) by at most 0.004 for either encoder.
- **Band width.** The pre-stated prediction fails. At matched capacity, widening the band to K=12 and K=24 narrows the hidden layer after the projection from 66 to 42 and 24 units. A further run therefore keeps K=24 at width 66 (79,638 encoder parameters). In no case does the B2 excess over raw samples shrink. Relative to K=6 it changes by a factor of 1.06 (matched) and 1.34 (wide) on plateaus, and 2.14 and 2.67 on Gaussians. Wider bands also raise the in-distribution validation error, from 0.025 to 0.033, 0.033 and 0.033. They raise the unseen-family error too, from 0.118 to 0.147, 0.161 and 0.155. This is consistent with the selection of K=6 on validation.
- **Code placement** (post hoc, not pre-registered). For every sweep IC we measure the distance from its IC code to the nearest training-IC code, in units of the RMS spread of the training codes (§5). At trained frequencies every encoder stays within 0.23 spreads. At sharp ICs the Fourier-feature codes move 1.5–2.6× as far as raw-sample codes (Table 7). Across all sweep points the error rises with this distance. The Spearman correlations are 0.74, 0.71, 0.80 and 0.89 for raw samples, factorized, spectral and LIFTSPEC.

Values outside the trained range (Fig. 4). Amplitudes up to 2× the trained maximum, and floors approaching zero, degrade all encoders gradually: 0.152 and 0.146 at a=2; 0.095 and 0.108 at c=0, where the logistic solution develops sharp fronts. Neither shows a cliff.

G Boundary-condition and multi-family studies: details

Design.

- **Boundary-condition study.** Convection with $\beta \in \{1..5\}$.
  - BC types: periodic, and inflow-Dirichlet $u(0,t) = g(t)$ with $g(t) = g_1 + (A + Bt)e^{-t/0.1}$, where $A = u_0(0) - g_1$ and $B = A/0.1 - \beta u_0'(0)$. This matches u_0 and its first time derivative at the corner. The boundary at $x = 2\pi$ is an outflow boundary and carries no data.
  - BC features: $[\mathbb{1}_{\text{periodic}}, \mathbb{1}_{\text{inflow}}, g_1]$. The raw-sample control appends them to its input, so its added capacity is $(64+3) \cdot 150$. The structured encoders match it.
  - Training (296 specifications): the zoo ICs under every BC configuration, plus each of the 128 trained G/S ICs once under a periodic BC and once under an inflow BC.
- **Multi-family study** (286 specifications). One model learns convection (both BC types) and reaction (periodic), with each trained IC appearing once in each family.

The exact lists are in `code/specification_encoder_plan_stretch.json`, frozen before any stretch run.

Table 8: **Boundary-condition study** (convection, $\beta \in \{1..5\}$, periodic and inflow-Dirichlet BCs; relative L_2, mean±std over seeds). Inflow references come from the authored solver of App. B.

Arm	seen	orig.	unseen BC cfg.	unseen params	unseen fam. (per.)	unseen fam. (inflow)	fam.×BC
Coef.-only P^{2}INN (ref. setting)	0.871±0.004	0.009±0.001	0.939±0.004	0.781±0.002	1.170±0.004	1.095±0.005	1.113±0.005
Coef.-only, IC-varied data	0.556±0.001	0.463±0.001	0.535±0.001	0.557±0.001	0.747±0.002	0.665±0.001	0.689±0.001
Raw-sample conditioning	0.045±0.002	0.024±0.002	0.182±0.020	0.231±0.014	0.331±0.021	0.333±0.025	0.332±0.023
factorized, raw	0.061±0.008	0.029±0.007	0.082±0.006	0.151±0.008	0.242±0.014	0.209±0.014	0.212±0.013
LIFTSPEC (lift–project, pre-registered)	0.047±0.003	0.023±0.003	0.097±0.020	0.209±0.033	0.400±0.028	0.363±0.036	0.363±0.035

Table 9: **One model for two PDE families** (convection with both BC types, and reaction; relative L_2, mean±std over seeds). The held-out type, convection–reaction, never appears in training.

Arm	orig. conv.	orig. reac.	conv. params	conv. families	reac. params	reac. families	held-out comp.	held-out type (CR)
Coef.-only, IC-varied data	0.455±0.001	0.259±0.001	0.587±0.000	0.709±0.001	0.349±0.000	0.517±0.002	0.628±0.001	0.453±0.006
Raw-sample conditioning	0.030±0.004	0.016±0.003	0.315±0.011	0.414±0.023	0.142±0.007	0.251±0.024	0.398±0.047	0.332±0.056
factorized, raw	0.038±0.005	0.019±0.001	0.226±0.012	0.288±0.012	0.103±0.008	0.181±0.020	0.261±0.019	0.321±0.047
LIFTSPEC (lift–project, pre-registered)	0.035±0.003	0.017±0.001	0.328±0.023	0.497±0.029	0.167±0.025	0.293±0.024	0.467±0.026	0.346±0.058

Table 10: **Breadth of a single model.** What one shared model is trained on, and how it does on specifications it never saw (relative L_2 of the factorized encoder, mean over seeds; from `logs/breadth_summary.json`). For the reaction model, the error range spans the never-trained families; for the PDE-family row, reaction / convection.

Axis	Trained jointly	Evaluated, never trained	Error	Reference
IC families (reaction)	2 trained: G, S	8 never trained: H, K, L, M, P, T, W, Z	0.060–0.110	0.324–0.689 (coef.-only)
BC types (convection)	2: periodic, inflow-Dirichlet ($g_1 \in \{0.1, 0.5, 0.9\}$)	inflow values $g_1 \in \{0.3, 0.7\}$	0.082	0.535 (IC-blind)
		never-trained families (P, T) under both BC types	0.209–0.242	$\geq$0.665 (IC-blind)
PDE families (one model)	2: convection, reaction (convection with both BC types)	unseen families in each	0.181 / 0.288	0.517 / 0.709 (IC-blind)
		held-out composition: convection, unseen family, unseen inflow value	0.261	0.628 (IC-blind)
		held-out PDE type: convection–reaction — *not covered*	0.321	0.317 / 0.377 (exact single mechanism)

Held-out PDE type: attempts. Convection–reaction ($\beta, \rho > 0$ together) never appears in training. We made three pre-registered attempts (Amendment 1).

- **Attempt 1, zero-shot.** The joint encoders reach 0.3463 (LIFTSPEC), 0.3315 (raw samples) and 0.3209 (factorized). The exact pure-convection and pure-reaction solutions score 0.317 and 0.377 against the convection–reaction reference (`logs/multi_cr/cr_single_mechanism.json`).
- **Attempt 2, term-wise coefficient branch.** The coefficient branch is replaced by $e_\beta(\beta)+e_\nu(\nu)+e_\rho(\rho)$, each term an MLP $1 \to 150 \to 150 \to 50$. This gives 101,544 encoder parameters, and the extra capacity is disclosed. Over 3 seeds the held-out type reaches 0.343 ± 0.052. Seen accuracy stays comparable: 0.033 and 0.018 on the original convection and reaction problems.
- **Attempt 3, code optimization (diagnostic).** For seeds 0–1 we optimize the 50-d code alone, with all weights frozen.
  - Fitting the reference on validation points (oracle) gives 0.070 for LIFTSPEC and 0.063 for raw samples, against 0.310 and 0.307 with the encoders' own codes.
  - Minimizing the physics loss gives 0.233 and 0.214.

  The decoder's solution manifold contains good approximations of convection–reaction solutions. The coefficient encoder cannot reach them for an unseen combination of terms, and physics-only refinement from that starting point stalls far from the oracle. A likely contributing cause is that the `modvec` path receives the raw, never-seen (β, ρ) pair.

H Attempts and deviations

- **Reference baseline seed collapse.** One of the five coefficient-only reference seeds is stuck in the reaction failure mode. We report it, include it in the pre-registered statistics, and add a stricter check that excludes its influence (§4).
- **First split draft (18 ICs).** It was replaced before the plan was frozen, because in-distribution validation showed insufficient IC coverage. Only validation points were used for that decision.
- **Full-length verbatim runs of the authors' scripts.** These exceeded our job time limit (App. A). We replaced them with trajectory and evaluation equivalence checks.
- **Round-1 negative result for the pre-registered encoder.** Reported in full. The follow-up encoder was tested only on sets defined after round 1.
- **Collapsed enlarged-budget run.** One of three $2\times$-budget LIFTSPEC runs did not fit its training set. It is excluded by a rule written after the collapse was seen, and the inclusive numbers are reported (App. F).
- **Falsified band-limit explanation of B2.** Our pre-stated explanation of the sharp-IC failure did not survive two band-width tests. We report the outcome and label the replacement explanation as post hoc (§6).

Details are in `attempts_log.md`.